COURSE INFORMATION

School/Faculty:	**Applied Computing & AI/Faculty Computing**			
Program name:	**Bachelor in Computer Science (Data Engineering) with Honours**			
Course code:	**SCST1133**	**Academic Session/Semester:**		**2026-2027/1**
Course name:	**Data Engineering Ecosystem**	**Pre/co requisite (course name and code, if applicable):**		
Credit hours:	**3**			

Course synopsis	This course introduces students to the field of infromation technology, data engineering by exploring core concepts, processes, tools, and the roles of Data Engineers, Data Scientists, and Analysts. Students will examine the data engineering ecosystem, including data structures, file formats, data sources, programming languages, and modern platforms. The course also covers Big Data technologies such as Hadoop and Spark, along with ETL/ELT processes, data pipelines, data lifecycle management, security, governance, ethics, and emerging career pathways. Teaching and learning activities include lectures, active learning, hands-on labs, industry talks, and industry visits, emphasising authentic work-based learning (WBL). By the end of the course, students will be able to analyze data engineering technologies using systems thinking, apply digital tools across the data lifecycle, and demonstrate ethical and responsible practices aligned with industry expectations.
Course coordinator (if applicable)	Dr. Aryati Bakri

Course lecturer(s)	**Name**	**Office**			**Contact no.**	**E-mail**
	Dr. Aryati Bakri	*N28 402-10*			*01159277751*	*aryati*
	Prof. Madya. Ts. Dr. Mohd Shahizan					*shahizan*

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	**[1]CLO**	**PLO [2](MQF Cluster Code)**	**3Taxonomies & 4Graduate attributes**	**5ESD, A & F**	**6VBE**	**T&L methods**	******Assessment methods**
CLO1	Explain basic concepts of information technology and their role in supporting data-driven systems through systems thinking.	PLO1 (C1)	C4 SC1		AT	Lecture, Active Learning, Industry Talk, Industry Visit	Asg,F
CLO2	Construct a basic data engineering life cycle by applying appropriate technologies, best practices, and industry standards.	PLO6 (C3D)	P3 A(IT2)		ST	Project Based Learning	Asg, PR, Pr
CLO3	Develop a prototype that addresses the data engineering ecosystem by applying the data engineering life cycle while demonstrating ethical principles and professional responsibilities as a data engineer.	PLO11 (C5)	A3 A(CC3)		N	Lecture, Active Learning, Industry Talk, Industry Visit	Asg, E

This is the basic mapping required for the CI. Any added information is allowed (extra columns for weight or other elements) provided this is made consistent for all CI at program/school/faculty level.

Refer **Taxonomies of Learning and ***UTM's Graduate Attributes for UG and Generic Skills for PG, where applicable for measurement of outcomes achievement
****Asg – Assignment;F –Final Exam; PR – Project; Pr – Presentation, E-E-portfolio

**MQF Cluster Code
C1 = Knowledge & Understanding, C2 = Cognitive Skills, C3A = Practical Skills, C3B = Interpersonal Skills, C3C= Communication Skills, C3D = Digital Skills, C3E = Numeracy Skills, C3F = Leadership, Autonomy & Responsibility, C4A = Personal Skills, C4B = Enterpreneurial Skills, C5 = Ethics & Professionalism

Details on Innovative T&L practices:

No.	Type	Implementation
1	Active learning	Active learning is implemented through collaborative such as gallery, flip classroom, problem-solving tasks, hands-on exercises, and guided discussions that require students to apply data engineering concepts in practical contexts
2	Project based Learning	Conducted based on given real world problem on design thinking such as the data driven solution on parking space in university. The students must manage their projects by submitting specified deliverables based on the given scenario.
3	Industry Talk	Share from industry on data engineering ecosystem in specific industry such as in PPG.
4	Industry Visit	Conducted the industry visit to observation on data engineering ecosystem in real industry.

Transferable skills (generic skills learning in course of study which can be useful and utilised in other settings) :

Engaging in ethical activities that create a positive impact.
Apply data engineering technology

Effective Learning Time (ELT) details:

Week/ Meeting	Course Content Outline and Subtopics	CLO*	Learning and Teaching Activities											TOTAL ELT
			Face-to-Face (F2F)									Non F2F Independent Learning		
			Physical			Online (Synchronous)						Online (Asynchronous)	Others	
			L	T	O	L	T	P	O					
Week 1	Topic 1: Introduction to Information Technology, Computer and System	CLO1				3						1	2	6
Week 2	Topic 2: Tools: Github, Video, Google Stitch, Figma	CLO1				3						1	2	6
Week 3	Topic 3: Data Collection & Design Thinking & Understanding the Data Landscape (Semi, Unstructured, Structured Data)	CLO1	3									1	2	7
Week 4	Topic 4: Introduction to Data Engineering	CLO1				3						1	2	6
Week 5	Topic 5: Data Engineering ecosystem & Lifecycle	CLO1	3									1	2	6
Week 6	**Industry Visit**	CLO1	3									2	4	9
Week 7	Topic 6: Career opportunities in the field of Data Engineering	CLO2				3						1	2	6
Week 8	**MID SEMESTER BREAK**													
Week 9	**Industry Talk 1**	CLO2	3									1	2	6
Week 10	Topic 7: Data Acquisition	CLO3	3									2	2	8
Week 11	**Industry Talk 2**	CLO3	3									2	2	8
Week 12	Topics 8: Data Governance & Ethics	CLO3				3						2	1	6
Week 13	Topic 9:Quantum computing in data engineering	CLO3	3									2	1	7
Week 14	Design Thinking: Project Presentation	CLO3	3									2	1	7
			21	0	0	9	0	0	0			**SUB-TOTAL ELT :**		**69**

				34				16	19	
	Continous Assessment	**%**	**Face-to-Face (F2F)**				**NF2F Independent Learning for Assessment**		**TOTAL ELT**
			Physical	**Online (Synchronous)**			**Online (Asynchronous)**	**Others**	
1	Project (Design Thinking & Figma) & Peer Assesment & Presentation	25	1				8	14	23
2	E-portfolio (Individu)	10					4	8	12
3	Assigment 1: Industry Talk * 3 - Report, Poster, News	15					5	9	14
4	Assigment 2: Industry Visit * 1 (News)	5					4	5	9
5	Google Video: Data Engineering Lifecycle	8					4	5	9
6	Quiz	7					1	1	2
							SUB-TOTAL SLT :		**69**
		70					**26**	**42**	

	Summative Assessment	**%**	**Face-to-Face (F2F)**				**NF2F Independent Learning for Assessment**		**TOTAL ELT**
			Physical	**Online (Synchronous)**			**Online (Asynchronous)**	Others	
1	Final Exam	30	3					9	12
							SUB-TOTAL ELT :		**12**
							ELT for Assessment:		**81**
							GRAND TOTAL ELT:		**150**

A	% SLT for F2F Physical Component [(Total F2F Physical)/(Total F2F Physical + Total F2F Online + Total Independent Learning) x 100)]		19.33
B	% SLT for Online & Independent Learning Component : [(Total F2F Online+Total Independent Learning)/(Total F2F Physical+Total F2F Online+ Total Independent Learning) x 100]		80.67
C	%SLT for Online Component: [(Total F2F Online+Total Online NF2F Independent Learning)/(Total F2F Physical+Total F2F Online+ Total Independent Learning) x 100]		16.67
D	% SLT for All Practical Component [% F2F Physical Practical + % F2F Online Practical]		2.67
D1	% SLT for F2F Physical Practical Component [(TotalF2F Physical Practical/(Total F2F Physical+ Total F2F Online+Total Independent Learning) x100)]		2.67
D2	% SLT for F2F Online Practical Component [Total F2F Online Practical/(Total F2F Physical + Total F2F Online + Total Independent Learning) x 100]		0.00
Please tick (/)if this course Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT) Note : ** For ODL Programme : Courses with mandatory practical requirement imposesd by programme standards or any related standards can be ecempted from complying to the minimum 80% ODL delivery rule inthe SLT.			

Identify special requirement or resources to deliver the course (e.g. software, nursery, computer lab, simulation room etc)

References (include required and further readings, and should be the most current)

Main references:
Praveen, R. V. S. (2024). Data Engineering for Modern Applications. Addition Publishing House.
Reis, J., & Housley, M. (2022). Fundamentals of data engineering. " O'Reilly Media, Inc.".
References
Trehan, A., & Pradhan, C. (2024). Automated data lineage tracking in data engineering ecosystem. International Research Journal of Modernization in Engineering Technology and Science, 6(12), 3305-3312.
LaDeau, S. L., Han, B. A., Rosi-Marshall, E. J., & Weathers, K. C. (2017). The next decade of big data in ecosystem science. Ecosystems, 20, 274-283.
Demchenko, Y., De Laat, C., & Membrey, P. (2014, May). Defining architecture components of the Big Data Ecosystem. In 2014 International conference on collaboration technologies and systems (CTS) (pp. 104-112). IEEE.
Curry, E., Metzger, A., Zillner, S., Pazzaglia, J. C., & García Robles, A. (2021). The elements of big data value: Foundations of the research and innovation ecosystem (p. 399). Springer Nature.

Other additional information (if applicable)

All students must adhere to the following to pass a computing course:

i. Pass both Continuous Assessment AND Final Assessment.

ii. Get at least minimum pass marks following UTM regulation (40% or D+).

The passing mark for:

- Continuous Assessment is 30 % of continuous marks.
- Final Assessment is 20% of the final marks.

If a student does not achieve the passing mark for Continuous Assessment OR the passing mark for the Final Assessment, then the student's overall final mark will be set below the UTM passing mark

Academic honesty and plagiarism: *(Below is just a sample)*

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES).
Copying of works (texts, lab results, etc.) from other students/groups or from other sources is not allowed. Brief quotations (texts in quotes) are allowed with the sources indicated. Existing texts should be paraphrased with your own words to explain what you understand. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Students who submit copied work will obtain zero for the assignment and exams and disciplinary steps may be taken by the faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (if applicable)

1. Attendance/participation is compulsory and will be taken during every lecture and meeting session. Students with less than 80% of total participation will be barred from presenting their project.
2. Students are required to behave and follow the dressing regulation and etiquette which has been stated in the University ruling while in class, in lab, and in exam hall*.
3. Any form of plagiarism is NOT ALLOWED. Students who are caught cheating for copying other students' assignments/lab exercises will get zero mark.
4. Exercises will be given in class and some may be taken for assessment. Students who do not take the exercise will lose the marks for the exercise.
5. Students who are absent will not be allowed to conduct the project demo and present the final assessments, except to those who are sick and submit medical certificates which are confirmed by UTM panel doctors. Make up demo can only be given within one week from the initial date of demo.
6. Iterative demo must be shown on the due dates. Some points will be deducted for the late demo only with lecturers' permission and accepted justification.

*Note: This classes are conduced hybrid as approved by the faculty/university until further notice.

Disclaimer:

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While every effort has been made to ensure the accuracy of the information supplied herein, Universiti Teknologi Malaysia cannot be held responsible for any errors or omissions.

ELT = (Theory + Industrial Guidance + Assessment) x 50%

Total of credit for LI/Practical = ELT/40 Notional Hours

Note: For ODL Programme : Courses with mandatory practical requirement imposed by programme standards or any related standards can be exempted from complying to the

Prepared by:	**Certified by:**
Name: Dr. Aryati Bakri	**Name:**

Signature: 	**Signature:**
Date: 11/10/2026	**Date:**